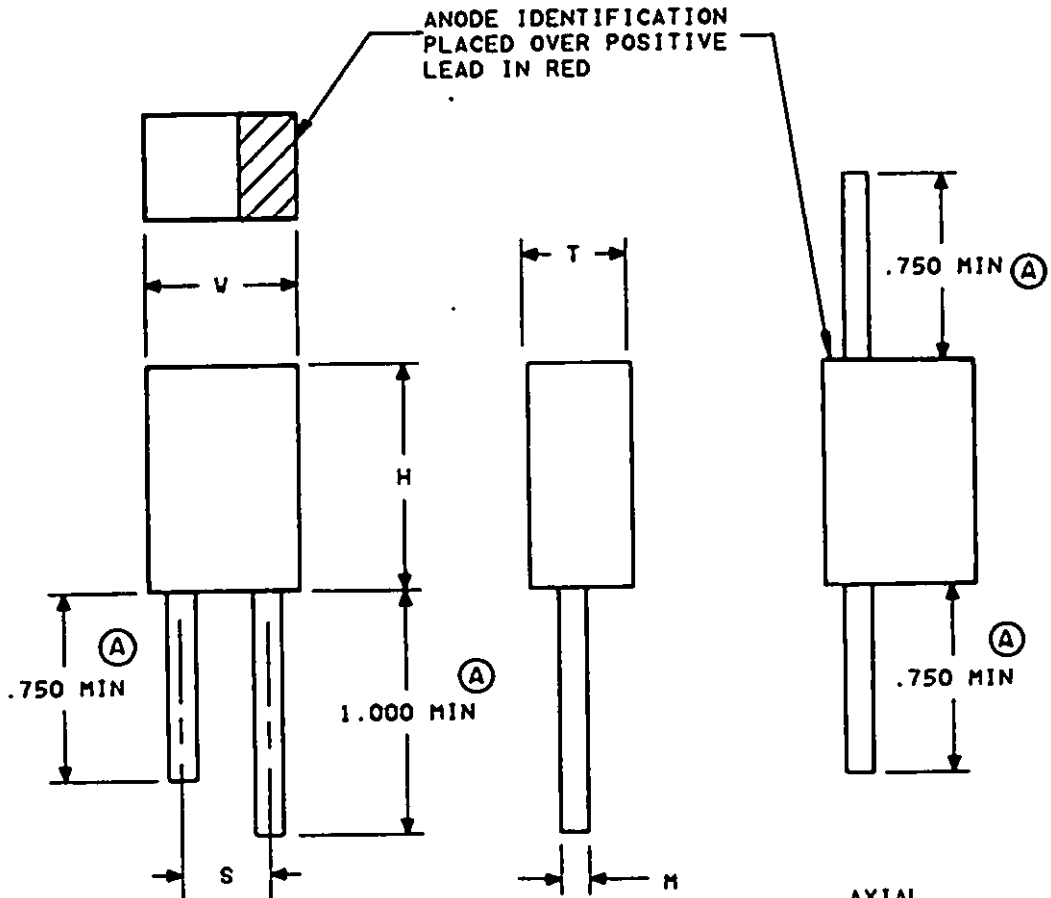
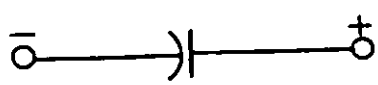
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (SOLID ELECTROLYTE), TANTALUM, POLAR, NONHERMETICALLY SEALED STYLES CX06 AND CX16

This specification sheet is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Military Specification MIL-C-49137.





CIRCUIT DIAGRAM

FIGURE 1. Styles CX06 and CX16 capacitors.

Ⓐ denotes changes

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FSC 5910

Case size	Dimensions (A)							
	T		W		H		M	S
	Min	Max	Min	Max	Min	Max	±.002	
A	.030	.040	.040	.050	.085	.100	.007	.030 ±.015
B	.030	.040	.060	.070	.110	.125	.010	.050 ±.015
C	.050	.070	.100	.120	.140	.165	.010	.100 ±.020
D	.060	.075	.160	.185	.195	.225	.010	.150 ±.020
E	.075	.110	.195	.220	.260	.290	.016	.180 ±.025
F	.100	.130	.200	.230	.280	.310	.016	.200 ±.025
G	.120	.150	.340	.375	.435	.475	.016	.300 ±.025

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.063	1.60	.160	4.06	.280	7.11
.007	.18	.070	1.78	.165	4.19	.290	7.37
.010	.25	.075	1.90	.180	4.57	.300	7.62
.015	.38	.085	2.16	.185	4.70	.310	7.87
.016	.41	.100	2.54	.195	4.95	.340	8.64
.020	.51	.110	2.79	.200	5.08	.375	9.52
.025	.64	.120	3.05	.220	5.59	.435	11.05
.030	.76	.125	3.18	.225	5.72	.475	12.07
.040	1.02	.130	3.30	.230	5.84	.750	19.05
.050	1.27	.140	3.56	.250	6.35	1.000	25.40
.060	1.52	.150	3.81	.260	6.60		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The positive lead on CX06 is at least .250 inch longer than the negative lead.

FIGURE 1. Styles CX06 and CX16 capacitor - Continued.

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TABLE I. Styles CX06 and CX16 characteristics.

Type designation 1/ (A)	DC rated voltage (@ 85°C)	Nominal capacitance	Dissipation factor (A) (max)	DC leakage (max) (A) 25°C	Case size
	Volts	µF	%	µA	
CX-6A474-	2	.47	10	.5	A
CX-6A225-	2	2.2	10	.5	B
CX-6A106-	2	10.0	10	.5	C
CX-6B155-	3	1.5	10	.5	B
CX-6B685-	3	6.8	10	.5	C
CX-6B226-	3	22.0	10	1.0	D
CX-6B227-	3	220.0	15	9.0	G
CX-6C334-	4	.33	10	.5	A
CX-6C105-	4	1.0	8	.5	B
CX-6C475-	4	4.7	8	.5	C
CX-6C156-	4	15.0	8	1.0	D
CX-6C476-	4	47.0	8	2.0	E
CX-6C686-	4	68.0	8	3.0	F
CX-6D224-	6	.22	10	.5	A
CX-6D684-	6	.68	6	.5	B
CX-6D335-	6	3.3	6	.5	C
CX-6D106-	6	10.0	6	1.0	D
CX-6D336-	6	33.0	6	2.0	E
CX-6D476-	6	47.0	6	3.0	F
CX-6D157-	6	150.0	10	9.0	G
CX-6F154-	10	.15	10	.5	A
CX-6F474-	10	.47	6	.5	B
CX-6F225-	10	2.2	6	.5	C
CX-6F685-	10	6.8	6	1.0	D
CX-6F226-	10	22.0	6	2.0	E
CX-6F336-	10	33.0	6	3.0	F
CX-6F107-	10	100.0	8	9.0	G
CX-6H104-	15	.10	10	.5	A
CX-6H334-	15	.33	6	.5	B
CX-6H155-	15	1.5	6	.5	C
CX-6H156-	15	15.0	6	2.0	E
CX-6H226-	15	22.0	6	3.0	F
CX-6H686-	15	68.0	8	9.0	G
CX-6J104-	20	.10	6	.5	B
CX-6J154-	20	.15	6	.5	B
CX-6J224-	20	.22	6	.5	B
CX-6J105-	20	1.0	6	.5	C
CX-6J335-	20	3.3	6	1.0	D
CX-6J475-	20	4.7	6	1.0	D
CX-6J106-	20	10.0	6	2.0	E
CX-6J156-	20	15.0	6	3.0	F
CX-6J476-	20	47.0	8	9.0	G
CX-6K684-	25	.68	6	.5	C
CX-6K225-	25	2.2	6	1.0	D
CX-6K685-	25	6.8	6	2.0	E
CX-6K106-	25	10.0	6	3.0	F
CX-6K336-	25	33.0	6	9.0	G
CX-6M104-	35	.10	6	.5	C
CX-6M154-	35	.15	6	.5	C
CX-6M224-	35	.22	6	.5	C
CX-6M334-	35	.33	6	.5	C
CX-6M474-	35	.47	6	.5	C
CX-6M684-	35	.68	6	1.0	D
CX-6M105-	35	1.0	6	1.0	D
CX-6M155-	35	1.5	6	1.0	D
CX-6M225-	35	2.2	6	2.0	E
CX-6M335-	35	3.3	6	2.0	E
CX-6M475-	35	4.7	6	2.0	E
CX-6M685-	35	6.8	6	3.0	F
CX-6M106-	35	10.0	6	9.0	G
CX-6M156-	35	15.0	6	9.0	G
CX-6M226-	35	22.0	6	9.0	G

1/ Complete type designation will include additional symbols to indicate style and capacitance tolerance.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Case material: Polyester sleeve and epoxy end filled.

Leads: Solder-coated metal.

DC rated voltage: See table I.

Operating temperature: -55° to +85°C.

DC leakage (DCL): See table I.

Capacitance (cap.): See table I.

(A) Cap. tolerance: ±10% (K) or ±20% (M).

Dissipation factor (DF): See table I.

(A) Terminal strength: In accordance with method 211 of MIL-STD-202.

Test condition A - 1/2 pound for case sizes A and B, 1 pound for case sizes C and D, and 2 pounds for case sizes E, F, and G.

Test condition C - 1/2 pound applied force - 3 bends.

Stability at low and high temperature: In accordance with MIL-C-49137.

Step 1 (+25°C):	DCL	-	See table I.
	Cap.	-	Within tolerance specified in table I.
	DF	-	See table I.
Step 2 (-55°C):	ΔCap.	-	Within -15 percent of step 1 measured value.
	DF	-	200 percent of table I.
Step 3 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.
Step 4 (+85°C):	DCL	-	10 times 25°C limit (see table I).
	ΔCap.	-	Within +15 percent of step 1 measured value.
	DF	-	150 percent of table I.
Step 5 (+25°C):	DCL	-	See table I.
	ΔCap.	-	Within 10 percent of step 1 measured value.
	DF	-	See table I.

Surge voltage: In accordance with MIL-C-49137.

DCL	-	See table I.
(A) ΔCap.	-	Within 10 percent of initial measured value.
DF	-	See table I.

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Life: Method 108 of MIL-STD-202.

1,000 hours:

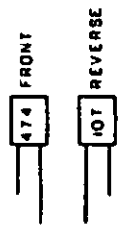
At 25°C: DCL - 200 percent of table I.

ΔCap. - Within 10 percent of initial measured value.

DF - See table I.

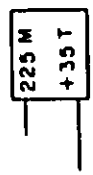
Marking: In accordance with MIL-C-49137 except that case sizes A and B shall be marked as shown in example 1. The front shall be marked with the capacitance value coded in picofarads and the reverse shall be marked with the dc voltage and the manufacturer's trademark. Case sizes E, F, and G shall be marked as shown in example 3. The first line will contain the part number and the second shall contain the date code (year and week), lot code, and manufacturer's trademark.

EXAMPLE 1



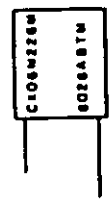
CASE SIZE
A and B

EXAMPLE 2



CASE SIZE
C and D

EXAMPLE 3



CASE SIZE
E, F and G

Voltage groups (85°C rated voltage):
I----------2 to 10V inclusive
II----------15 to 35V inclusive

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-A356)

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